
AGENTCRAFT
NATIONAL AI AGENT CHALLENGE
BUILD INTELLIGENT AGENTS FOR A BETTER TOMORROW
Innovate
Collaborate
Solve Real-World Problems
Create Impact
SAME WORLD BRIGHTER TOMORROW
7 TRACKS
REAL IMPACT
AI AGENTS
SUSTAINABLE FUTURE
ABOUT THE EVENT
AgentCraft is a national-level AI Agent Challenge that brings together innovators, builders and problem-solvers to design autonomous AI agents for real-world impact. Set in a Minecraft-inspired world, participants will tackle critical global challenges aligned with the United Nations Sustainable Development Goals (SDGs).
OUR VISION
To empower the next generation of AI builders to create autonomous solutions for a more inclusive, sustainable and resilient world.
THEME
A Minecraft-inspired world where AI agents build solutions for people, the planet and a brighter tomorrow.
THE 7 PROBLEM TRACKS
SAME WORLD. BIGGER IMPACT.
Each track is designed with real-world complexity, requiring autonomous AI agents that can plan, reason, use tools, adapt to dynamic situations and deliver measurable outcomes.
01 VidyaMitra
Quality Education (SDG 4)
Personalized learning, intelligent tutoring, and learning-gap identification.
KEY REQUIREMENTS
Knowledge graph with 30+ concepts
Adaptive learning paths
Misconception detection
Persistent learner memory
Assessment and improvement tracking
Difficulty: Very High
02 KrishiMitra
Zero Hunger (SDG 2)
AI-powered agriculture, smart irrigation, and sustainable farming.
KEY REQUIREMENTS
Multi-source farm data ingestion
25+ field zones with varied conditions
Irrigation planning under constraints
Forecast-aware decision making
Impact measurement (water, yield, efficiency)
Difficulty: Very High
03 AarogyaSetu AI
Good Health and Well-being (SDG 3)
Preventive health support and access to reliable health information.
KEY REQUIREMENTS
Evidence-grounded health guidance
Context-aware follow-up questions
Emergency escalation logic
Source verification and safety rules
Privacy-preserving design
Difficulty: Very High
04 GreenGuardian
Responsible Consumption and Production (SDG 12)
Intelligent waste management and sustainable resource utilization.
KEY REQUIREMENTS
Waste classification (10+ classes)
Recovery and recycling planning
Route optimization (30+ locations)
Resource tracking and audit trail
Dynamic replanning for disruptions
Difficulty: Very High
05 RakshaGrid
Disaster Intelligence and Response (SDG 11 & 13)
Autonomous emergency coordination and resource allocation.
KEY REQUIREMENTS
Simulation with 100+ affected locations
Multi-agent rescue and logistics
Dynamic route planning
Live replanning under disruptions
Fair and effective resource allocation
Difficulty: Extremely High
06 GridMind
Intelligent Energy Management (SDG 7)
Renewable energy forecasting, microgrid optimization, and autonomous load management.
KEY REQUIREMENTS
Grid with 20+ buildings and generators
Demand and renewable forecasting
Battery and load optimization
Constraint-aware scheduling
Reliable and cost-efficient energy supply
Difficulty: Extremely High
07 CyberSentinel
Autonomous Cyber Defense (SDG 9)
Threat detection, incident investigation, and safe automated defense.
KEY REQUIREMENTS
Multi-source log ingestion
Anomaly detection and correlation
Automated incident investigation
Authorized response and rollback
Evidence preservation and audit trail
Difficulty: Extremely High
COMMON ENGINEERING REQUIREMENTS (APPLY TO ALL TRACKS)
Genuine autonomy with multi-step planning
Memory and state management
Multi-agent collaboration
Fault tolerance and safe actions
Reproducible evaluation with baseline comparison
EVALUATION RUBRIC (100 POINTS)
Agent autonomy and planning 20
Technical depth and architecture 20
Reliability and safety 20
Measurable performance 20
Innovation and real-world impact 10
Demonstration and reproducibility 10
WHAT TO SUBMIT
Solution presentation and impact analysis
Runnable implementation (code + setup)
Agent architecture and workflow
Evaluation report with baseline comparison
Live or recorded demonstration
Responsible AI statement

OFFICIAL PROBLEM TRACKS & TECHNICAL REQUIREMENTS

A Minecraft-inspired national challenge where teams build autonomous AI agents that plan, use tools, collaborate, adapt to changing environments, and demonstrate measurable real-world impact.

Challenge principle: Build more than a chatbot. Build an agent that acts responsibly, verifies outcomes, recovers from failure, and proves its impact.

Seven SDG-aligned tracks • Common agent-engineering requirements • Reproducible evaluation

The AgentCraft Challenge

The Minecraft-inspired theme is more than a visual skin. Teams may represent farms, villages, hospitals, recycling facilities, disaster zones, energy networks, and digital infrastructure as a simulated block world. Agents interact through defined tools or APIs and must respond to changing conditions.

What makes this a difficult agent challenge?

- Autonomous multi-step execution: agents select tools, verify outputs, and revise plans instead of following a fixed script.
- Persistent memory: agents maintain task-relevant state and learn from prior actions or failures.
- Multi-agent coordination: at least two specialized agents communicate through defined interfaces and resolve conflicts.
- Fault tolerance: systems handle missing inputs, tool failures, stale data, and unexpected changes.
- Evidence-based evaluation: organizers rerun held-out scenarios and compare results against a simpler baseline.

Final seven tracks

Track	Focus	SDG alignment
01 VidyaMitra	Adaptive learning and knowledge-gap detection	SDG 4
02 KrishiMitra	Precision agriculture and smart irrigation	SDG 2
03 AarogyaSetu AI	Preventive health information and safe navigation	SDG 3
04 GreenGuardian	Waste recovery and circular resource management	SDG 12
05 RakshaGrid	Disaster response and emergency coordination	SDGs 11, 13
06 GridMind	Renewable energy and microgrid optimization	SDG 7
07 CyberSentinel	Defensive cybersecurity and incident response	SDG 9

Track requirements are intended as measurable prototype targets. Teams should state assumptions, document data provenance, and demonstrate their results in a controlled environment.

TRACK 01

VidyaMitra — Quality Education

SDG 4: Quality Education

PROBLEM STATEMENT

Build an autonomous AI learning agent that identifies a student's hidden knowledge gaps, creates a personalized learning strategy, adapts to performance, and verifies whether learning has occurred. It must understand what a learner knows and what they should learn next—not simply answer questions.

MANDATORY REQUIREMENTS

- Represent at least 30 concepts, prerequisite relationships, and common misconceptions in one selected subject.
- Maintain per-concept mastery estimates using answer correctness, difficulty, history, and confidence where available.
- Generate and revise learning paths instead of following a fixed lesson sequence.
- Offer hints, examples, and stepwise explanations before revealing a final answer.
- Retain learning history across sessions and update mastery estimates as new evidence arrives.
- Detect at least five predefined misconception patterns and select targeted corrective activities.
- Generate or select new assessment questions without simply repeating previously seen questions.
- Measure learning improvement using held-out questions, not only satisfaction ratings or response quality.

ADVANCED DIFFICULTY REQUIREMENTS

- Distinguish a knowledge gap from a careless mistake using diagnostic questions.
- Handle learners who perform well on easy questions but fail prerequisite concepts.
- Revise the teaching strategy when the initial approach fails or evidence becomes contradictory.
- Compare against a fixed learning-path baseline on the same evaluation cohort.

DIFFICULTY	Very high

EXPECTED DELIVERABLE: A working tutor, knowledge graph, learner-progress dashboard, reproducible test dataset, and evaluation report.

TRACK 02

KrishiMitra — Zero Hunger

SDG 2: Zero Hunger

PROBLEM STATEMENT

Build an autonomous agricultural agent that manages a simulated farm using soil conditions, weather forecasts, crop growth stages, and limited water. It must decide when, where, and how much to irrigate while protecting crop health and adapting to changing conditions.

MANDATORY REQUIREMENTS

- Process soil moisture, soil temperature, ambient temperature, humidity, rainfall, crop stage, and weather forecast data.
- Represent at least 25 farm zones with different soil properties, water needs, and crop conditions.
- Detect missing, delayed, noisy, and implausible sensor readings.
- Produce zone-specific watering schedules under a fixed water budget and irrigation-system capacity.
- Use crop type, growth stage, and estimated water demand in decisions.
- Revise plans when rainfall forecasts or observed weather change.

- Track prior irrigation actions and resulting soil-condition changes.
- Report water consumption, crop-stress indicators, irrigation efficiency, and resource use against a baseline.

ADVANCED DIFFICULTY REQUIREMENTS

- Handle at least three simultaneous disruptions, such as sensor failure, unexpected rainfall, and water shortage.
- Distinguish faulty moisture sensors from genuine dry soil using neighboring readings and historical patterns.
- Prevent repeated or conflicting watering actions from multiple zone agents.
- Demonstrate better water efficiency without exceeding predefined crop-stress limits in a controlled simulation.
- Explain which measurements, forecasts, and constraints drove each decision.

DIFFICULTY	Very high

EXPECTED DELIVERABLE: A farm simulation or replayable dataset, irrigation agent, field-zone dashboard, fault-handling demonstration, and quantitative impact report.

TRACK 03

AarogyaSetu AI — Good Health and Well-being

SDG 3: Good Health and Well-being

PROBLEM STATEMENT

Build a reliable health-information agent that helps users navigate preventive health information, understand general guidance, and identify when professional or urgent medical assistance is appropriate. The system must prioritize evidence, uncertainty, and escalation rather than unsupported diagnosis.

MANDATORY REQUIREMENTS

- Use an approved corpus of trusted health information with source metadata and dates.
- Cite passages supporting important recommendations.
- Ask relevant follow-up questions when essential context is missing.
- Provide general information about health awareness, screening, and prevention within the chosen scope.
- Identify predefined emergency warning signs and direct users toward appropriate urgent care.
- Abstain or request clarification when information is insufficient or sources conflict.
- Minimize personal data collection and avoid retaining unnecessary sensitive information.
- Evaluate answer accuracy, citation correctness, escalation behavior, and unsafe-answer frequency.

ADVANCED DIFFICULTY REQUIREMENTS

- Detect outdated, irrelevant, or inconsistent retrieved documents.
- Resist prompts that attempt to override medical safety rules.
- Distinguish general health education from situations requiring professional assessment.
- Evaluate unseen cases with incomplete context, misleading claims, and conflicting evidence.
- Verify that citations are real and support the claims made.

DIFFICULTY	Very high

EXPECTED DELIVERABLE: A health-information assistant, searchable evidence base, source-linked responses, safety test suite, and evaluation report.

SAFETY BOUNDARY: Use synthetic cases and trusted public information. Do not build a system for autonomous diagnosis or unsupervised treatment.

TRACK 04

GreenGuardian — Responsible Consumption and Production

SDG 12: Responsible Consumption and Production

PROBLEM STATEMENT

Build an autonomous circular-resource agent that identifies waste streams, selects suitable recovery pathways, coordinates collection, and optimizes reuse and recycling under operational constraints.

MANDATORY REQUIREMENTS

- Classify at least 10 predefined waste classes using images, structured records, or both.
- Recognize ambiguous, contaminated, and mixed-material waste.
- Choose among reuse, recycling, composting, and authorized disposal pathways.
- Schedule collection across at least 30 pickup locations under vehicle-capacity and time constraints.
- Track quantities entering and leaving each processing stage.
- Coordinate sorting, collection, and facility-allocation agents without double-counting resources.
- Respond to full bins, facility shutdowns, traffic delays, and unexpected waste volumes.
- Measure recovery rates, contamination, transport cost, landfill diversion, and material losses.

ADVANCED DIFFICULTY REQUIREMENTS

- Prevent unsuitable or contaminated materials from entering incompatible processing streams.
- Enforce mass balance: every unit of material must have a traceable disposition.
- Recalculate routes when facilities become unavailable without violating vehicle capacity.
- Compare against a baseline collection strategy under identical conditions.
- Maintain an audit trail for each batch, including classification confidence and destination.

DIFFICULTY	Very high

EXPECTED DELIVERABLE: A waste-management simulator, classification and planning agents, traceable material ledger, operational dashboard, and impact evaluation.

TRACK 05

RakshaGrid — Disaster Intelligence and Response

SDGs 11 and 13: Sustainable Cities and Climate Action

PROBLEM STATEMENT

Develop a multi-agent disaster-response system that coordinates evacuation, rescue assignments, emergency supplies, and shelter allocation in a simulated region affected by floods or another defined disaster. Conditions change while the system operates, so agents must replan with incomplete information.

MANDATORY REQUIREMENTS

- Represent at least 100 affected locations, 10 rescue units, five shelters, and multiple supply depots.
- Prioritize locations using severity, vulnerability, accessibility, and report reliability.
- Plan routes while accounting for road closures, travel times, and changing accessibility.
- Allocate rescue units, medical supplies, and shelter capacity without exceeding available resources.

- Include distinct reconnaissance, rescue, logistics, and shelter-management responsibilities.
- Update plans when incidents, blocked roads, or resource shortages appear.
- Handle delayed, missing, and contradictory reports from simulated field agents.
- Record why each rescue or allocation decision was made and which constraints were considered.

ADVANCED DIFFICULTY REQUIREMENTS

- Inject at least five unforeseen events, including a critical route closure, resource shortage, and contradictory reports.
- Prevent multiple agents from assigning the same rescue unit to incompatible tasks.
- Balance urgency and fairness rather than always prioritizing the nearest locations.
- Handle communication outages without allowing agents to act indefinitely on stale information.
- Compare response time, unmet critical needs, resource utilization, and allocation fairness against a baseline.

DIFFICULTY	Extremely high

EXPECTED DELIVERABLE: A disaster-world simulator, autonomous response agents, live operational map, event replay, and quantitative evaluation.

SAFETY BOUNDARY: All rescue decisions must be tested in simulation, not used for operational emergency dispatch.

TRACK 06

GridMind — Intelligent Energy Management

SDG 7: Affordable and Clean Energy

PROBLEM STATEMENT

Build an energy-management agent that coordinates renewable generation, battery storage, building demand, and critical loads in a simulated microgrid. It must minimize operating cost and unmet demand while respecting physical constraints and adapting to uncertain forecasts.

MANDATORY REQUIREMENTS

- Model at least 20 buildings, three renewable generators, five storage units, and critical and non-critical loads.
- Forecast energy demand from historical consumption and contextual features.
- Estimate solar or wind generation under changing weather conditions.
- Optimize battery charging and discharging within capacity, efficiency, and charge-rate limits.
- Prioritize available energy according to declared criticality and service requirements.
- Coordinate generation, storage, demand, and grid-management agents using shared state or a defined protocol.
- Prevent infeasible dispatch plans, battery overcharging, and allocation beyond available capacity.
- Track operating cost, renewable utilization, peak demand, unmet load, and critical-service reliability.

ADVANCED DIFFICULTY REQUIREMENTS

- Evaluate forecast errors, demand spikes, generator outages, and storage degradation.
- Use a rolling planning horizon to account for future demand.
- Compare against both a fixed-rule controller and a defined optimization baseline.
- Recover safely from infeasible schedules or stale data.
- Explain trade-offs among cost, reliability, renewable utilization, and critical-load protection.

DIFFICULTY	Extremely high

EXPECTED DELIVERABLE: A microgrid simulator, energy-management agents, real-time dashboard, baseline comparisons, and reproducible scenario results.

SAFETY BOUNDARY: Use a simulated grid with no connection to physical infrastructure or live power-control systems.

TRACK 07

CyberSentinel — Autonomous Cyber Defense

SDG 9: Industry, Innovation and Infrastructure

PROBLEM STATEMENT

Develop a defensive AI-agent system that monitors synthetic system logs, identifies suspicious behavior, investigates incidents across data sources, and recommends or executes authorized containment actions in an isolated sandbox.

MANDATORY REQUIREMENTS

- Process at least three synthetic data sources, such as authentication events, application logs, and network-flow records.
- Identify suspicious activity using rules, statistical methods, machine learning, or a justified combination.
- Correlate related events into incident timelines rather than isolated alerts.
- Gather evidence, assess confidence, and produce a defensible incident explanation.
- Recommend or execute predefined defensive actions, such as isolating a simulated endpoint or revoking a test credential.
- Require authorization for disruptive actions and provide rollback.
- Record source events, reasoning summaries, selected actions, and resulting state.
- Measure detection rate, false-positive rate, time to detection, investigation quality, and response correctness on held-out scenarios.

ADVANCED DIFFICULTY REQUIREMENTS

- Handle previously unseen combinations of suspicious events.
- Distinguish malicious activity from legitimate behavior that looks unusual.
- Resist prompt injection in untrusted logs, tickets, or retrieved incident notes.
- Detect unsupported or conflicting conclusions across investigation agents.
- Recover from failed response actions, unavailable tools, and duplicate alerts.
- Avoid unnecessary containment when evidence is weak.

DIFFICULTY	Extremely high

EXPECTED DELIVERABLE: An isolated cyber range, defensive agent architecture, incident dashboard, synthetic scenarios, auditable response engine, and evaluation report.

SAFETY BOUNDARY: Use synthetic data and authorized isolated systems only. No real-world exploitation, unauthorized scanning, or destructive actions are required.

Shared Engineering Requirements

These requirements apply to all seven tracks and are mandatory for eligibility.

1. Genuine autonomy

Complete a multi-step task, select tools based on the current state, verify outputs, and change plans when needed. A fixed sequence of LLM calls is insufficient.

2. Memory and state

Maintain persistent task-relevant state, record prior actions, and avoid repeating completed work. Demonstrate state updates and recovery after failure.

3. Multi-agent collaboration

Implement at least two agents with distinct responsibilities, explicit interfaces, and a conflict-resolution strategy. Explain why the multi-agent design adds value.

4. Fault tolerance and safe actions

Handle missing inputs, failed tools, timeouts, and unexpected environment changes. Consequential actions must obey track-specific constraints and approval rules.

5. Reproducible evaluation

Evaluate against a simpler baseline on at least 30 held-out scenarios. Report task success, failure modes, resource use, and track-specific metrics. Organizers should rerun tests independently.

Recommended evaluation protocol

- Publish problem statements, APIs, starter data, baseline instructions, and safety rules before development.
- Keep some final evaluation scenarios undisclosed while ensuring they remain within the published problem scope.
- Include at least one failure or recovery case in the final demonstration.
- Require teams to disclose model choices, external services, data sources, limitations, and evaluation assumptions.
- Judge the outcome and reliability—not the number of agents, size of the model, or polish of the interface alone.

Official Judging Rubric

Category	Points	What judges evaluate
Agent autonomy and planning	20	Multi-step execution, tool selection, replanning
Technical depth and architecture	20	Algorithms, data flow, coordination, engineering quality
Reliability and safety	20	Failure recovery, constraints, uncertainty, safe behavior
Measurable performance	20	Baseline comparisons, held-out tests, track-specific outcomes
Innovation and real-world impact	10	Novelty, feasibility, SDG relevance
Demonstration and reproducibility	10	Working demo, documentation, repeatable results
TOTAL	100	All categories scored against published criteria

Recommended judging rules

- A polished UI cannot compensate for an agent that fails its core task.
- Unsupported claims of accuracy or impact do not earn performance points.
- Organizers may introduce an undisclosed scenario during the final demonstration.
- A simpler, more reliable architecture can outperform a more complicated one.
- Safety-critical failures may cap or disqualify a submission according to rules published before the competition.

Participant Submission Checklist

- **Solution presentation:** problem definition, users, novelty, architecture, workflow, results, limitations, and SDG impact.
- **Runnable implementation:** source code, setup instructions, dependencies, and reproducible execution command.
- **Agent architecture:** roles, tools, memory design, state transitions, and coordination strategy.
- **Evaluation package:** baseline, held-out test results, failure analysis, and metric calculations.
- **Demonstration:** live or recorded execution showing successful operation and at least one failure or recovery scenario.
- **Responsible AI statement:** data sources, privacy safeguards, limitations, and safety boundaries.

Organizer note

The numerical targets in this document are proposed competition requirements. Organizers should publish the final APIs, dataset formats, test harness, resource limits, deadlines, and disqualification rules before registration closes. Provide starter datasets and a baseline so difficulty comes from engineering quality rather than unequal access to compute.

AGENTCRAFT • BUILD YOUR AGENT. SHAPE YOUR WORLD.